

AnchorSlice: Anchor-Guided Static Slicing for Token-Efficient Smart Contract Vulnerability Detection

SAJAD AGHANASIRI¹, SINA ZAKER¹, ALIREZA SHAMELI-SENDI¹

¹Faculty of Computer Science and Engineering, Shahid Beheshti University (SBU), Tehran, Iran (e-mail: {s.aghanasiri, a_shameli}@sbu.ac.ir)

Corresponding author: Alireza Shameli-Sendi (e-mail: a_shameli@sbu.ac.ir).

This work was not supported by any specific grant but was conducted as part of academic research at Shahid Beheshti University.

ABSTRACT Automated vulnerability detection is a prerequisite for deploying smart contracts safely, because a deployed contract is immutable and every defect in it remains reachable by an adversary. Detectors that reason directly over source code have become increasingly capable, but their cost, latency and context capacity are bounded by the number of tokens they read, and a typical Solidity contract is dominated by comments, library code and boilerplate that cannot host a vulnerability. This paper presents AnchorSlice, a static slicing stage that makes smart contract vulnerability detection token-efficient by reducing each contract, before it reaches the detector, to the fragments capable of hosting each vulnerability category of the DASP taxonomy. AnchorSlice is purely lexical: category-specific anchor expressions and brace matching decide what survives, overlapping regions are merged once and tagged with every category they may host, and unmodified library constructs are collapsed to stubs under a rescue rule that preserves evidence. The slicer needs no compilation and processes a contract in 7.6 ms on average. On a stratified sample of 50 contracts from the DIVE benchmark, it retains 60.2% of source characters while preserving evidence for all 119 labelled category instances. In a paired evaluation in which only the detector input changes, slicing reduced token consumption by 56.6% and detection runtime by 19.7%, while recall rose from 47.9% to 63.0%, F1-score from 42.7% to 50.0% and Cohen's kappa from 0.145 to 0.220. The detection gains are not statistically significant (McNemar exact test, $p = 0.761$), but the results show that taxonomy-aware input reduction can more than halve the token cost of smart contract vulnerability detection without degrading it.

INDEX TERMS smart contract vulnerability detection, token efficiency, static slicing, input reduction, blockchain security, DASP taxonomy

I. INTRODUCTION

DETECTING vulnerabilities in smart contracts before they are deployed is one of the few effective defences available on public blockchains, and detectors that read contract source code directly have become considerably more capable in recent years. Their cost, however, grows with the length of what they read, while much of a deployed contract is irrelevant to any vulnerability. This section motivates the problem of *token-efficient smart contract vulnerability detection*. It first explains why contract defects are costly and why existing analysers leave the detection problem open, then shows how the input length of source-level detectors has become a limiting factor, and finally introduces AnchorSlice, summarises the contributions of this work, and poses the research questions that structure its evaluation.

Smart contracts turn contractual clauses into code that

executes without a trusted intermediary [1], and on Ethereum they hold and move assets directly [2]. That property is also the problem. A contract is immutable once deployed, its bytecode is public, and any defect in it is permanently reachable by an adversary who can read the source. The 2016 DAO incident, in which a reentrancy fault drained roughly sixty million dollars, made this concrete and motivated the first generation of automated analysers [3]. Practitioners subsequently organised the recurring fault patterns into the Decentralized Application Security Project (DASP) Top 10 [4], which remains the vocabulary against which most detection work is evaluated.

A decade of tooling has not settled the problem. When nine widely used analysers were run over 47,587 deployed contracts, they collectively identified only a minority of the annotated faults and flagged 97% of the corpus as vulnera-

ble, a false-positive rate that makes unattended use impractical [5]. Controlled bug injection reaches the same conclusion from the opposite direction: injected faults of well-understood shapes go undetected by tools designed specifically to find them [6]. The difficulty is not that analysers such as Slither [7] reason poorly about control flow, but that many real defects are semantic. Whether a missing modifier matters depends on what the function does, and no fixed data-flow pattern captures that. Learning-based detectors were introduced partly to close this gap [8].

Detectors that reason over source code with pre-trained language models changed the trade-off again. Because such models carry general knowledge of programming idioms and security reasoning, a contract can be examined directly, without a hand-written rule for each fault class. Pipelines that pair model reasoning with static confirmation [9], that split auditing into generation and criticism [10], or that fine-tune a detector and then argue about its output through cooperating agents [11] all report detection quality beyond that of pattern-based tools, and a recent systematic review counts a rapidly growing body of such work [12]. In all of these systems the language model is a component of the detector rather than the object of study; what they share is that the detector consumes the contract source.

This dependence has a cost that scales with input length. A source-level detector is billed per token and bounded by a context window, and long inputs also degrade reasoning: accuracy falls when the relevant evidence sits in the middle of a long input rather than near its edges [13]. For smart contracts the waste is easy to see. A fixed-supply ERC-20 token is several hundred lines of library boilerplate wrapped around perhaps thirty lines that could plausibly hold a flaw, yet the whole file is submitted. Prompt compression attacks the general version of this problem by using a small model to drop low-information tokens [14], but that approach is domain-agnostic, itself consumes model capacity, and offers no guarantee that security-relevant code survives.

We take the opposite route and exploit the fact that vulnerability classes have lexical footprints. A reentrancy fault requires an external call, a bad-randomness fault requires a block-derived value, and an arithmetic fault requires an operator. Code containing none of these cannot host the corresponding defect, and deciding that requires no model. This paper introduces AnchorSlice, a purely static slicing stage that keeps, for each DASP category, only the regions whose tokens make the category possible, and hands the detector a compact annotated slice instead of the full file. The design borrows the intuition of program slicing [15] but deliberately stops short of dependence analysis, so that the slicer runs in milliseconds and never needs to compile. It extends the candidate-pruning idea of our earlier LineGuard framework [16], which restricts the detector's view for one vulnerability category at a time, to all eight categories in a single pass with a measurable evidence guarantee. We evaluate AnchorSlice on the DIVE multi-label benchmark [17], whose 22,330 contracts are annotated for the eight DASP categories

we target.

The primary contributions of this paper are summarised as follows:

- We present *AnchorSlice*, a detector-agnostic, purely lexical slicing stage for token-efficient smart contract vulnerability detection. It defines an anchor set for DASP categories 1–8, a visibility gate, a tag-preserving merge that emits each source region once while labelling it with every category it may host, and a library-collapse step with a rescue rule that restores lost evidence.
- We validate the slicer against the DIVE benchmark and show that it retains evidence for all 119 labelled category instances in our sample while emitting 60.2% of the original characters at an average of 7.6 ms per contract. We also report which semantic restrictions are compatible with the benchmark's annotation policy and which are not.
- We report a paired evaluation in which only the detector input changes. Slicing cut token consumption by 56.6% and detection runtime by 19.7% while raising recall, F1-score and chance-corrected agreement, and we test the significance of these changes and analyse which anchors carried useful information and which misled the detector.

Our study is structured around the following research questions:

- **RQ1 (Evidence retention):** To what extent can a purely lexical, taxonomy-anchored slice reduce the source of a smart contract while preserving the code evidence for every labelled vulnerability category?
- **RQ2 (Token efficiency):** How much do the token consumption, runtime and monetary cost of vulnerability detection fall when the detector receives the slice instead of the complete contract?
- **RQ3 (Detection effectiveness):** How does slicing affect vulnerability detection effectiveness, both in aggregate and for each DASP category?

The remainder of this paper is structured as follows. Section II reviews previous work on smart contract vulnerability detection and input reduction. Section III introduces AnchorSlice and its design. Section IV presents the experimental evaluation and answers the research questions. Section V discusses limitations and threats to validity. Finally, Section VI concludes the paper and outlines future work.

II. BACKGROUND AND RELATED WORK

Token-efficient vulnerability detection draws on bodies of work that have rarely been considered together: analysers and detectors for smart contract vulnerabilities, techniques that reduce the input an analysis component must process, and the labelled benchmarks against which both are evaluated. This section reviews them in that order. Sections II-A to II-C cover rule-based program analysis, learning-based detection, and source-level detectors built on language models, with attention to how much input each consumes. Section II-D

reviews input reduction and program slicing, Section II-E the benchmarks used for evaluation, and Section II-F positions AnchorSlice against the closest prior work, as summarised in Table 1.

A. RULE-BASED SECURITY ANALYSIS

Oyente introduced symbolic execution of EVM bytecode and defined several of the fault patterns still in use [3]. Securify derived compliance and violation patterns from dependence graphs to make its verdicts explainable [18], while SmartCheck worked on an intermediate XML representation of Solidity to keep analysis lightweight [19]. Mythril combined symbolic execution with taint analysis over bytecode [20], and MAIAN targeted trace-level properties such as contracts that can be made to leak or lock funds [21]. VeriSmart pushed precision further for arithmetic safety by inferring transaction invariants [22]. Dynamic approaches complement these: ContractFuzzer generated transaction sequences against an instrumented EVM [23], and sFuzz added an adaptive strategy for branches that random input rarely reaches [24]. Slither occupies a middle ground, converting Solidity to an SSA-based intermediate form that supports fast detectors and code-review aids [7]. The large-scale comparisons of these tools [5] and the injection studies that probe them [6] both conclude that coverage remains partial and precision low, which is the gap learning-based and model-based detectors have tried to close. Because rule-based tools execute fixed analyses rather than reading tokens, input length is not their bottleneck; it becomes one only for the detectors reviewed next.

B. LEARNING-BASED VULNERABILITY DETECTION

VulDeePecker showed that vulnerability detection could be learned from code gadgets, slices of semantically related statements gathered around library and API calls [25]. SySeVR generalised the idea to four syntactic seed types and used program dependence graphs for slicing [26]. Devign moved to graph representations that combine abstract syntax, control flow and data flow [27], and LineVul demonstrated that transformer models can localise predictions to individual lines [28], often building on code-specific pre-training such as CodeBERT [29]. In the contract setting, Zhuang et al. represented functions as normalised graphs and applied a degree-free graph convolutional network and a temporal message propagation network [8]; later work combined such graph models with expert-designed patterns to improve interpretability [30]. AnchorSlice is related to this literature mainly through the seed-and-expand idea: like VulDeePecker and SySeVR it begins from syntactic anchors, but it produces a reduced source for a general-purpose detector rather than a feature vector for a trained classifier, and it therefore needs no dependence graph.

C. LANGUAGE-MODEL DETECTORS AND THEIR INPUT COST

Code-trained models [31], together with in-context learning [32] and chain-of-thought prompting [33], made it practical to obtain a security judgement from a general model. GPTScan uses model reasoning to nominate candidate variables and statements and dedicated static modules to validate them, reporting recall above 70% on ground-truth logic bugs [9]. GPTLens splits the task between an auditor role that proposes vulnerabilities and a critic role that scores them, which suppresses over-reporting [10]. LLM4Vuln decouples the components of vulnerability reasoning so that knowledge retrieval, tool use and reasoning can be measured separately [34]. LLM-SmartAudit organises multiple agents around a conversation about the contract [35], and SmartLLaMA post-trains an open model in two stages to produce both a decision and an explanation [36]. iAudit fine-tunes a detector and a reasoner and then has ranker and critic agents debate the cause, reaching an F1-score of 91.21% on a curated set of 263 vulnerabilities [11]. PropertyGPT goes further and generates formal properties for verification rather than verdicts alone [37]. What these detectors share is the assumption that they receive the contract, and the systematic review of the area identifies context length as a recurring constraint [12]. Our earlier framework, LineGuard [16], already narrows the detector's view: for a vulnerability category given in advance, it ranks candidate lines with category-specific regular expressions and passes the top-ranked candidates with a fixed context window. That pruning serves line-level localisation of one category per query, and its effect on evidence retention and token cost was not isolated. AnchorSlice is orthogonal to all of these detectors, since any of them could consume a sliced contract instead of a whole one.

D. INPUT REDUCTION AND PROGRAM SLICING

The cost of long inputs has been attacked generically. LLM-Lingua uses a small language model to identify and drop low-information tokens, reporting high compression with limited loss [14], and LongLLMLingua adds question-aware reordering for long-context settings [38]. Retrieval-augmented generation avoids the problem by fetching only relevant passages [39]. These methods estimate relevance statistically and therefore cannot promise that a particular security-critical line survives. Classical program slicing offers the opposite guarantee, computing the statements that can affect a criterion [15], and thin slicing restricts the result to statements that produce the value of interest [40]. Full slicing needs a compilable program and a dependence graph, which is a strong requirement for a corpus of deployed contracts spanning many compiler versions. AnchorSlice sits between these positions: it is coarser than dependence-based slicing, but its slicing criterion is the vulnerability taxonomy itself, and its evidence retention against a labelled benchmark can be measured directly.

TABLE 1: Positioning of AnchorSlice Against Prior Work on Vulnerability Detection and Input Reduction

Work	Year	Technique	Input to the detector	Input reduction	Model-free reduction	Taxonomy-guided
[3] Oyente	2016	Symbolic execution	EVM bytecode	None	–	–
[18] Securify	2018	Dependence-graph patterns	EVM bytecode	None	–	–
[19] SmartCheck	2018	XML-based patterns	Solidity source	None	–	–
[7] Slither	2019	SSA-based static analysis	Solidity source	None	–	–
[25] VulDeePecker	2018	Deep learning	Code gadgets	Slicing around API calls	✓	X
[26] SySeVR	2022	Deep learning	Semantic slices	Dependence-based slicing	✓	X
[8] Zhuang et al.	2020	Graph neural network	Contract graph	Graph normalisation	✓	X
[40] Thin slicing	2007	Program slicing	Program	Value-flow slicing	✓	X
[14] LLMingua	2023	Prompt compression	Compressed prompt	Small-model token dropping	X	X
[38] LongLLMLingua	2024	Prompt compression	Compressed prompt	Question-aware token dropping	X	X
[16] LineGuard	2025	Model-based line localisation	Pruned candidate lines	Per-category regex pruning	✓	Single category
AnchorSlice	2026	Anchor-guided static slicing	Tagged multi-category slice	Anchored windows, library collapse	✓	✓

E. LABELLED BENCHMARKS

Evaluation in this area depends on the quality of labels. ScrawlID labelled real contracts by majority vote over five analysers [41], and DAppSCAN built large source and bytecode datasets from 1,199 audit reports covering 682 DApp projects [42]. We use DIVE [17], which annotates 22,330 contracts deployed between 2016 and 2024 against the eight DASP categories. DIVE aggregates MAIAN, Mythril, Semgrep, Slither, Solhint and VeriSmart through a power-based voting scheme and then re-checks positive findings against category-specific code evidence, a post-hoc validation that corrected about 14.3% of DoS and 24.9% of time-manipulation labels. Labels are assigned at contract level, which shapes how evidence retention can be validated.

F. POSITIONING OF ANCHORSlice

Table 1 contrasts AnchorSlice with the closest prior work along the dimensions that matter for token-efficient detection. Rule-based analysers do not reduce their input because they do not read tokens. Learning-based detectors reduce input through slicing or graph construction, but their reduction is tied to a trained representation rather than to a vulnerability taxonomy. Prompt-compression methods reduce input for token-bound models but use a model to decide what to keep and offer no guarantee for security-relevant code. LineGuard prunes with category-specific rules but for one category at a time. AnchorSlice is, to our knowledge, the first reduction stage that is simultaneously model-free, guided by the full vulnerability taxonomy, and validated for evidence retention on a labelled smart contract benchmark.

III. ANCHORSlice

AnchorSlice turns a complete Solidity contract into a compact, annotated slice that retains, for every DASP category, the code regions able to host that category, so that a downstream vulnerability detector reads fewer tokens without losing the evidence it needs. Fig. 1 places the slicer in the detection pipeline, and Algorithm 1 gives the complete procedure, which runs in three stages: comment removal, library collapse, and anchor-guided window extraction. This section first states the principles that constrain the design (Section III-A), then describes the lexical anchors and the single semantic gate retained (Section III-B), the extraction

Algorithm 1 AnchorSlice

Require: Source S , categories $\mathcal{C}$ with anchors $\{A_c\}$, gated set $\mathcal{V} \subseteq \mathcal{C}$, cap κ , padding π

Ensure: Annotated slice Σ

- 1: $L \leftarrow \text{StripComments}(S)$
- 2: $K \leftarrow \text{lines of top-level library units in } L \text{ \{collapsed\}}$
- 3: $W \leftarrow \emptyset$
- 4: **for** each function $f = (f_s, f_e, sig)$ in L with $f_s \notin K$ **do**
- 5: $G \leftarrow \mathcal{V}$ **if** sig is `view/pure` **else** $\emptyset$
- 6: **for** $k = f_s$ **to** f_e **do**
- 7: $T \leftarrow \{c \in \mathcal{C} \setminus G : A_c \text{ matches } L[k]\}$
- 8: **if** $T \neq \emptyset$ **then**
- 9: $(a, b) \leftarrow \text{InnermostBlock}(L, f, k, \kappa, \pi)$
- 10: $W \leftarrow W \cup \{(a, b, T)\}$
- 11: **end if**
- 12: **end for**
- 13: **end for**
- 14: $B \leftarrow \text{MergeAdjacent}(W) \text{ \{union of tags\}}$
- 15: **for** each $c \in \mathcal{C}$ tagged in no block of B **do**
- 16: **if** some $k \in K$ has A_c matching $L[k]$ **then**
- 17: $B \leftarrow B \cup \{(k-1, k+1, \{c\})\} \text{ \{rescue\}}$
- 18: **end if**
- 19: **end for**
- 20: $\Sigma \leftarrow \text{Render}(\text{stubs}(K), \text{declarations}(L), B)$
- 21: **if** $|\Sigma| \geq |L|$ **then**
- 22: $\Sigma \leftarrow L \text{ \{small contract: send whole\}}$
- 23: **end if**
- 24: **return** Σ

and tag-preserving merge of windows (Section III-C), library collapse and its evidence rescue rule (Section III-D), and finally how the slice is rendered and handed to the detector (Section III-E).

A. DESIGN PRINCIPLES

Three requirements shaped the design. First, the slicer must be fully static, in the sense that every decision follows from regular expressions and brace counting; a slicer that needed model judgement to decide what to keep would spend the token budget it is meant to save. Second, it must emit each region of source at most once. Running eight independent category-specific passes would be the obvious construction,

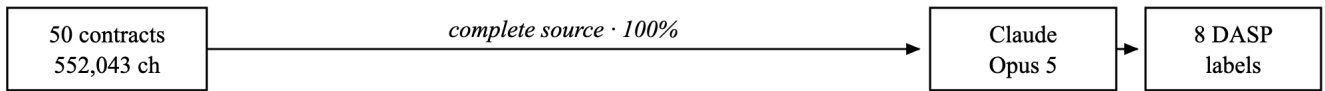
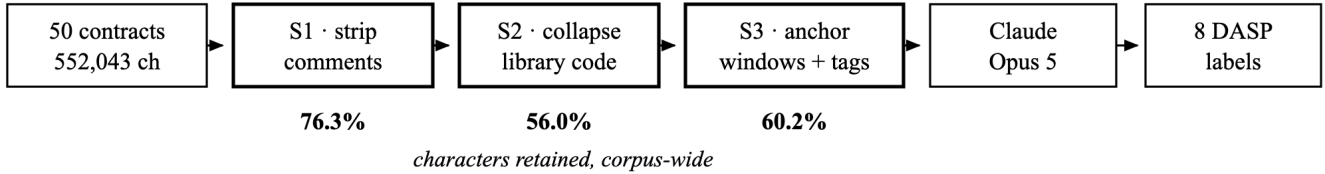
Arm A — baseline**Arm B — AnchorSlice**

FIGURE 1: Vulnerability detection with and without AnchorSlice. Arm A passes the complete source to the detector; Arm B inserts the three AnchorSlice stages S1–S3, annotated with the share of corpus characters retained after each. The detector (Claude Opus 5 in this study), its prompt, and its output schema are identical in both arms.

but the passes overlap heavily and their union would exceed the original file, so AnchorSlice performs a single pass and attaches a set of candidate categories to each emitted region (Algorithm 1, lines 4–14). Third, the slicer must be detector-agnostic: its output is ordinary Solidity text with comment headers, so any source-level detector can consume it without modification.

B. LEXICAL ANCHORS AND THE VISIBILITY GATE

Each category owns a set of anchor expressions, and a line matching one marks its enclosing region as a candidate for that category (line 7). Two macros carry most of the weight. `EXT` matches any external or cross-contract call, including `low-level call`, `send`, `transfer`, `delegatecall` and `staticcall` forms, calls through an interface cast, and calls on a receiver that is not a language builtin. `MUTFN` matches any public or external function that is not declared `view` or `pure`. The second macro deserves comment. Access-control faults are often an absence rather than a presence, so no positive pattern can find them; treating every state-mutating entry point as a candidate is the only static way to surface a contract whose fault is a missing modifier.

We evaluated five semantic gates intended to suppress obviously safe regions and retained one. Suppressing arithmetic candidates in contracts compiled under Solidity 0.8, where overflow reverts by default, cut arithmetic recall from 1.00 to 0.43, because DIVE labels many such contracts positive regardless. Requiring a state write before admitting a reentrancy candidate cut its recall from 0.91 to 0.36. Treating a call wrapped in `require` as a checked return cut that category from 0.60 to 0.20. Only the visibility gate survived: a `view` or `pure` body cannot mutate state, and excluding such bodies from the five state-dependent categories cost no recall at all (line 5). The rejected gates were sound as Solidity reasoning but disagreed with the benchmark’s annotation policy, which is a useful reminder that a slicer tuned to a dataset inherits that

dataset’s conventions. Table 2 gives the complete anchor set, the gate applied to each category, and the evidence retention the resulting slicer achieves.

C. WINDOW EXTRACTION AND TAG-PRESERVING MERGE

An anchored line is expanded to the innermost brace block that contains it, capped at $\kappa = 10$ lines; past that cap the window falls back to the anchor with $\pi = 2$ lines of context on each side (line 9). The enclosing function signature is attached separately so that visibility and modifiers stay visible even when the body is truncated. Windows are then sorted and merged: any two whose ranges overlap or abut become a single block whose tag set is the union of theirs (line 14). A region relevant to both reentrancy and arithmetic is consequently emitted once and carries both labels, which preserves the single-emission principle. Across the evaluation corpus this produces 14.9 blocks per contract.

D. LIBRARY COLLAPSE AND EVIDENCE RESCUE

Unmodified reference implementations account for a large share of a typical file. AnchorSlice detects top-level `SafeMath`, `Address`, `Context`, `Ownable`, `ERC20` and common exchange interfaces by name and replaces each with a one-line stub, so the detector still knows the construct is present (line 2). Applied naively, this cost recall, dropping access control to 0.95 and DoS to 0.86, because in a few contracts the only anchor for a category lay inside a collapsed construct. A rescue pass repairs the loss: if a category ends the pass with no block anywhere in the file while an anchor for it exists inside collapsed text, a three-line window is emitted at that anchor (lines 15–19). The rescue restores recall to 1.00 at a cost of about two percentage points of size.

E. SLICE RENDERING AND DETECTOR INTERFACE

The emitted slice begins with the stubs for collapsed libraries and the contract-level declarations, then lists the blocks in

TABLE 2: Anchor Specification for the Eight DASP Categories

Category	Anchor expressions	Gate	Retained
Reentrancy	EXT	view/pure	22/22
Access Control	modifier· only\w+· (·)owner =· tx.origin· selfdestruct· suicide· delegatecall· require(msg.sender· renounce· transferOwnership· authorize· admin· MUTFN	view/pure	39/39
Arithmetic	[+ - * / %]=·++·--·x [+ * / %] y·x - y·**·<<·>>·unchecked	–	23/23
Unchecked Return Values	EXT	view/pure	5/5
DoS	for(·while(·.push(·.length·require(·revert·assert(·EXT	view/pure	7/7
Bad Randomness	blockhash· block.{difficulty, coinbase, gaslimit, number, timestamp, prevrandao}· now· keccak256· sha3(· sha256· random· nonce· seed	–	5/5
Front Running	approve(· allowance· msg.value· tx.gasprice· price· bid· swap*(· slippage· amountOutMin· deadline· reserve· MUTFN	view/pure	5/5
Time manipulation	now· block.timestamp· block.number· N {days, hours, minutes, seconds, weeks}· deadline· startTime· endTime· cooldown· lastClaim· timeout	–	13/13
All categories	–	–	119/119

EXT matches low-level calls (.call, .call.value, .send, .transfer, delegatecall, staticcall), calls through an interface cast, transferFrom/_transfer, swap and add-liquidity calls, payable (conversions, inline assembly, and calls on any receiver that is not a language builtin. MUTFN matches any public or external function not declared view or pure. The gate column names the only semantic restriction retained after ablation. *Retained* counts the label-positive contracts that keep at least one block tagged with the category.

source order (line 20). Each block carries a header naming its original line range and its candidate categories, and line numbers refer to the original file so that a finding can be traced back. Fig. 2 shows an excerpt of the slice produced for a contract from the evaluation sample. The header reports that 61 of 128 source lines were kept in five blocks, and SafeMath is reduced to a stub. Block 2 is a modifier tagged for access control because it constrains msg.sender and for DoS because it contains require; block 3 is a public constructor-style function tagged by MUTFN for access control and front running. A short preamble tells the detector that these tags are lexical hints rather than findings and that most tagged blocks are not vulnerable. A guard handles small files: when the block headers would cost more than the removed lines save, the contract is sent whole with comments stripped (lines 21–23), which occurred for 11 of the 50 contracts in our sample.

IV. EXPERIMENTAL EVALUATION

This section evaluates whether AnchorSlice makes smart contract vulnerability detection more token-efficient without degrading it. The evaluation holds the detector fixed and changes only its input, comparing the complete contract (Arm A) with the AnchorSlice output (Arm B) on the same contracts. Section IV-A describes the experimental setup and configuration, Section IV-B the dataset and sampling procedure, and Section IV-C the evaluation protocol and metrics. Sections IV-D to IV-G report evidence retention, token consumption and runtime, detection effectiveness, and per-category behaviour. Section IV-H revisits the research questions posed in Section I.

A. SETUP

AnchorSlice and the evaluation harness are implemented in Python using only the standard library, and the com-

```
// SLICED CONTRACT 19288: 128 src lines -> 61 kept (48%),
// 5 candidate blocks.
// Each block lists the DASP categories whose static
// anchors fired inside it.
// Omitted code matched no anchor for any category.
// [standard SafeMath omitted - unmodified reference
// implementation]
// --- DECLARATIONS ---
pragma solidity ^0.4.18;
[...]
// --- BLOCK 2 | src L16-21 | candidates: Access Control,
// DoS
// modifier onlyCLevel() {
//   require(
//     msg.sender == ceoAddress
//   );
// }
// --- BLOCK 3 | src L23-26 | candidates: Access Control,
// Front Running
// function AdPotato() public{
//   ceoAddress=msg.sender;
//   initialize(0
//     x58AFF91f5b48245Bd83deeB2C7d31875f68b3f0D);
// }
// [...]
```

FIGURE 2: Excerpt of the AnchorSlice output for DIVE contract 19288. Lines marked [...] are elided here for space; all other lines are reproduced verbatim.

plete source, sample definitions, raw results and run logs are available in the project repository [43]. Slicer runtime was measured on an Intel Core i9-9880H processor with Python 3.12.3. The detector in both arms is Claude Opus 5 at high reasoning effort, accessed through the Claude Code command-line interface in non-interactive print mode. All built-in tools and external tool servers are disabled and no session state is persisted, so the detector sees nothing beyond the prompt and cannot retrieve the answer from elsewhere. A JSON schema constrains every reply to eight binary labels with a confidence and a short justification per category, which

TABLE 3: Experimental Configuration and Parameters

Parameter	Value	Description
<i>Slicer</i>		
Window cap κ	10 lines	Largest brace block kept whole
Padding π	± 2 lines	Context around an anchor past the cap
Gated categories $\mathcal{V}$	5	Skipped inside <code>view/pure</code> bodies
<i>Sampling</i>		
Sample size	50	Seeded stratified sample, seed 20260827
Category floor	5	Minimum positives and negatives per category
Size filter	1.2–30 kB	Excludes empty stubs, bounds per-call cost
<i>Detector</i>		
Model	Opus 5	Claude Opus 5, high reasoning effort
Access	Claude CLI	Print mode, tools and tool servers disabled
Output	JSON schema	8 binary labels, confidences, reasons
System prompt	calibrated	Category definitions with examples and decision-threshold guidance
Calls	1 per contract	Sequential, 900 s timeout
Sampling	not exposed	Temperature and top-p not user-controllable

removes output parsing as a source of error. Each contract receives a single call, calls are issued sequentially, and the interface reports token usage, latency and cost for every call. The interface does not expose sampling parameters such as temperature, so the results are reproducible in configuration and in the logged per-call record but not bit-for-bit. Table 3 lists every parameter used; all of them are held fixed across both arms.

B. DATASET AND SAMPLING

We drew 50 contracts from DIVE [17] using a seeded random sample that was repaired so that every category has at least five positive and five negative instances. This keeps base rates close to the corpus prevalence while making all eight categories measurable in both directions; a sample composed only of heavily labelled contracts would leave precision undefined, because a detector answering all-positive would score perfectly. Contracts smaller than 1,200 bytes were excluded as empty stubs, and contracts larger than 30,000 bytes were excluded to bound the cost of each call. The sample totals 552,043 characters, with a median contract of 11,484 characters and a maximum of 27,687, so no contract required truncation. Compiler versions split 27 legacy contracts against 23 at Solidity 0.8 or above. The 400 label cells (50 contracts $\times$ 8 categories) contain 119 positives, a positive rate of 29.8%.

C. EVALUATION PROTOCOL

Arm A submits the complete source of each contract. Arm B submits the AnchorSlice output preceded by a short preamble explaining the block format and stating that the category

tags are lexical hints, not findings. Everything else is held constant: the same detector, reasoning effort, system prompt, output schema and contracts, with one call each. The system prompt defines each DASP category with a minimal code example and calibrates the detector’s decision threshold to the benchmark, instructing it to mark a category when the vulnerable pattern is present and not demonstrably mitigated. Because both arms classify the same 400 cells, the comparison is paired, and we report McNemar’s exact test on discordant cells alongside the aggregate measures.

We report label agreement over all cells, Hamming loss, exact match at contract level, and micro-averaged precision, recall and F1-score. Because the label distribution is unbalanced, we also report Cohen’s kappa, which corrects for agreement expected by chance. Resource use is measured as total tokens across all token classes, wall-clock time, and cost. Evidence retention is measured without the detector, as the share of label-positive contracts that keep at least one block tagged with the corresponding category. All detection metrics use the DIVE labels as the reference.

D. EVIDENCE RETENTION AND SLICE COMPACTNESS

AnchorSlice reduced the sample from 552,043 to 332,134 characters, or 60.2% of the original, with a mean per-contract reduction of 34.0%. Evidence retention was 1.00 for every category: all 119 label-positive category instances kept at least one block tagged with that category, as summarised in Table 2. Decomposing the pipeline in Fig. 1 shows that comment removal brings the corpus to 76.3% and library collapse to 56.0%, with the anchoring stage returning 60.2% after its block headers are counted. The slicer processed the 50 contracts in 378.5 ms, a mean of 7.6 ms and a maximum of 17.5 ms per contract, averaged over five passes, which is negligible beside the tens of seconds a detector call takes.

E. TOKEN CONSUMPTION, RUNTIME AND COST

Table 4 reports resource use. Slicing cut total token consumption from 913,958 to 396,438, a reduction of 56.6%, and mean consumption per contract from 18,279 to 7,928 tokens. Wall-clock time fell from 1,019.6 to 818.5 seconds, or 19.7%, which matters for throughput when detection is scaled beyond a sample; in both arms the interface-reported inference time accounted for essentially all of the wall-clock time. Output tokens fell by 15.6%, a smaller proportion than the input reduction, because the length of the detector’s reasoning is governed by the eight decisions it must make rather than by how much source it reads. Cost decreased by only 1.4%, for reasons discussed in Section V.

F. DETECTION EFFECTIVENESS

Detection quality improved on every aggregate measure, as Table 5 shows. Recall rose from 47.9% to 63.0%, an absolute gain of 15.1 percentage points, and precision rose from 38.5% to 41.4%, so the additional detections were not obtained by indiscriminate over-reporting. F1-score improved from 42.7% to 50.0%, and Cohen’s kappa from 0.145 to 0.220,

TABLE 4: Resource Consumption Across 50 Contracts

Measure	Arm A	Arm B	Change
Input tokens	204	102	−50.0%
Output tokens	63,798	53,872	−15.6%
Cache-write tokens	125,385	176,438	+40.7%
Cache-read tokens	724,571	166,026	−77.1%
Total tokens	913,958	396,438	−56.6%
Tokens per contract	18,279	7,928	−56.6%
Wall clock (s)	1,019.6	818.5	−19.7%
Seconds per contract	20.4	16.4	−19.7%
Cost (USD)	3.4103	3.3636	−1.4%

TABLE 5: Detection Effectiveness Over 400 Label Cells

Metric	Arm A	Arm B	Change
Label agreement	61.8%	62.5%	+0.7 pp
Hamming loss	38.2%	37.5%	−0.8 pp
Exact match (8/8)	0/50	2/50	+2
Precision	38.5%	41.4%	+2.9 pp
Recall	47.9%	63.0%	+15.1 pp
F1-score	42.7%	50.0%	+7.3 pp
Cohen’s kappa	0.145	0.220	+0.075
True positives	57	75	+18
False positives	91	106	+15
False negatives	62	44	−18

a relative gain of 51.7%. Label agreement rose from 61.8% to 62.5%, and Hamming loss fell correspondingly. Two contracts were classified correctly in all eight categories under slicing, whereas the baseline achieved none.

The confusion matrix locates the change. False negatives fell from 62 to 44 while true positives rose from 57 to 75, against an increase in false positives from 91 to 106. At contract level, 14 contracts improved, 23 were unchanged and 13 regressed. The gain is therefore concentrated in recovering vulnerabilities the baseline missed rather than in eliminating spurious reports. These aggregate improvements are not statistically significant: over the 400 paired cells, 20 cells were correct only under the baseline and 23 only under slicing, giving a two-sided McNemar exact p of 0.761.

G. PER-CATEGORY BEHAVIOUR

Table 6 reports every category, and Fig. 3 shows the three that improved. Access control gained 14.0 percentage points, by far the largest movement, and is also the category in which the baseline was weakest, marking only 6 of 39 positives. The MUTFN anchor tags precisely the state-mutating entry points where a missing modifier would live, and under slicing the detector identified 11 more, cutting false negatives from 33 to 24 at a cost of two false positives. Front running gained 6.0 points and eliminated both of its false negatives, and reentrancy gained 4.0 points while reducing false negatives from four to one. Unchecked return values and time manipulation held their agreement, and the former halved its false negatives, so six of the eight categories either improved or held. Arithmetic and bad randomness each lost 2.0 points, and DoS lost 14.0 points, which we attribute to an over-broad anchor discussed

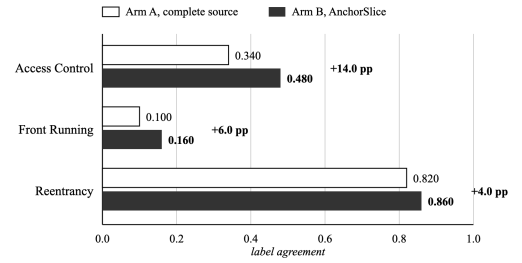


FIGURE 3: Label agreement with and without slicing for the three categories in which agreement improved. Table 6 reports all eight categories.

in Section V.

H. DISCUSSION

The central observation is that discarding roughly 40% of the source characters did not degrade the detector’s judgement and, on aggregate measures, improved it. The simplest reading is that the removed text was not being used. Comment blocks, licence headers and unmodified library implementations carry no information on which a security judgement depends, and the detector appears to treat them accordingly. This is consistent with the observation that evidence buried inside long inputs is attended to less reliably than evidence near the edges [13]: shortening the input raises the density of security-relevant material and shortens the distance between related fragments. The stage decomposition adds a second insight. Comment removal and library collapse account for essentially all of the size reduction, while the anchoring stage removes 22.3% of the remaining source and adds 16.9% back as block headers. The value of anchoring therefore lies less in compression than in localisation: the tags tell the detector where each category could plausibly live, which suggests that an anchor is useful in proportion to how selective it is.

We now revisit the three research questions raised in Section I.

RQ1 — Evidence retention. AnchorSlice retained 60.2% of the characters of the 50-contract sample, with 14.9 blocks per contract, and every one of the 119 label-positive category instances kept at least one block tagged with its category. Comment removal and library collapse delivered most of the reduction, and the rescue rule was necessary to reach full retention after collapse. The slicer needs no compilation and processes a contract in 7.6 ms on average.

To best answer RQ1: A purely lexical, taxonomy-anchored slice can remove about two fifths of a contract’s source while preserving evidence for every labelled vulnerability category in our sample, at a runtime cost that is negligible within a detection pipeline. Because retention is measured at contract level and the anchors were developed on the same sample, this result establishes that the slicer does not discard labelled evidence in-sample, not that it retains the specific vulnerable statements on unseen contracts.

RQ2 — Token efficiency. Slicing reduced total token consumption by 56.6%, from 18,279 to 7,928 tokens per contract,

TABLE 6: Per-Category Detection Detail Across All Eight DASP Categories

Category	DIVE +	Detector + (A)	Detector + (B)	Agreement (A)	Agreement (B)	Change (pp)	FN (A)	FN (B)
Reentrancy	22	23	27	0.820	0.860	+4.0	4	1
Access Control	39	6	17	0.340	0.480	+14.0	33	24
Arithmetic	23	17	22	0.560	0.540	-2.0	14	12
Unchecked Return Values	5	12	16	0.700	0.700	0.0	4	2
DoS	7	13	20	0.760	0.620	-14.0	3	3
Bad Randomness	5	3	4	0.960	0.940	-2.0	2	2
Front Running	5	46	47	0.100	0.160	+6.0	2	0
Time manipulation	13	28	28	0.700	0.700	0.0	0	0
All categories	119	148	181	0.618	0.625	+0.7	62	44

DIVE + counts reference positives; Detector + counts contracts the detector marked positive; FN counts false negatives. Change is the shift in label agreement in percentage points.

and wall-clock detection time by 19.7%. The reduction fell mainly on cache-read input tokens, which dropped by 77.1%, while output tokens fell by 15.6%, and monetary cost fell by only 1.4% because the removed tokens belong to the cheapest token class.

To best answer RQ2: AnchorSlice more than halves the token consumption of source-level vulnerability detection and shortens its runtime by about a fifth. Token count and latency are the portable benefits; the monetary saving depends on how a detector's pricing weights cached input against output and was small in our configuration.

RQ3 — Detection effectiveness. Under slicing, recall rose by 15.1 percentage points, precision by 2.9, F1-score by 7.3 and Cohen's kappa from 0.145 to 0.220, with false negatives falling from 62 to 44. Access control, front running and reentrancy improved; unchecked return values and time manipulation held; arithmetic and bad randomness declined slightly; and DoS declined by 14.0 points because its anchor fires almost everywhere. The paired McNemar test does not reach significance ($p = 0.761$).

To best answer RQ3: Slicing did not harm vulnerability detection while more than halving token use, and it improved every aggregate effectiveness measure in our sample, driven by recovered false negatives in categories whose anchors are selective. Because the improvements are not statistically significant and each arm was run once, we read them as encouraging rather than established, and the firm conclusion is that taxonomy-aware input reduction is compatible with effective detection. The DoS result identifies anchor selectivity as the property to control in future versions of the slicer.

V. LIMITATIONS AND THREATS TO VALIDITY

The results above come from one detector, one benchmark and a single run per arm, and several properties of the design and of the evaluation bound how far they generalise. This section states these limits explicitly so that the reported numbers are interpreted within them. It first addresses the strength of the detection results and the gap between token and monetary savings, then two design issues exposed by the per-category analysis, and finally threats arising from how evidence retention was validated, from the sample, and from the detector used.

Statistical strength. The aggregate detection improvements are not statistically significant. McNemar's exact test over the 400 paired cells gives 20 cells correct only under the baseline against 23 correct only under slicing, for a two-sided p of 0.761. The gains in recall and F1-score should therefore be read as encouraging rather than established; the claim the experiment supports firmly is that slicing does not harm detection while more than halving token use.

Token savings versus cost savings. The monetary saving is much smaller than the token saving. Cost fell by 1.4% against a 56.6% token reduction because the reduction falls almost entirely on cached input, the cheapest token class, while output tokens changed comparatively little. Detectors whose pricing weights these classes differently will see a different figure, so token count and latency are the more portable benefits.

An over-broad anchor. One anchor is clearly mis-specified. The DoS anchor includes `require`, `revert` and `assert`, which appear in nearly every non-trivial function, so almost every block carries a DoS tag. The detector read that ubiquity as evidence and raised its positive count from 13 to 20 against a reference of 7, without recovering any false negative. A tag that fires almost everywhere carries no information yet still shifts the detector's prior. Relatedly, library collapse identifies constructs by name, so a contract that reuses a standard name for a modified implementation would have that code collapsed as well.

Confounded treatment. Arm B both removes code and adds category tags, and the DoS result suggests that these two changes can act in opposite directions, so the aggregate outcome may understate the benefit of removal alone. A tag-free arm is needed to separate them.

Weak retention criterion. DIVE labels contracts rather than lines, so evidence retention of 1.00 means only that a positive contract keeps some block tagged with that category, not that it keeps the specific statements an annotator had in mind. The DIVE labels are themselves derived from a consensus of static analysers with post-hoc validation, so effectiveness is measured against the benchmark's annotation policy rather than against audited ground truth.

In-sample development and single runs. The anchors were developed by iterating against the same 50 contracts used for

evaluation, so the retention figure is an in-sample fit and will be optimistic on held-out data. The sample is also small, and each arm was run once, so part of the discordance between arms reflects the detector's own non-determinism.

One detector. Both arms use a single proprietary model accessed through an interface that does not expose sampling parameters or allow a dated model snapshot to be pinned. Although the slice is detector-agnostic by construction, its effect on other source-level detectors, including open-weight models and multi-stage pipelines such as LineGuard [16], has not been measured.

VI. CONCLUSION AND FUTURE WORK

This paper addressed the token cost of smart contract vulnerability detection and showed that a large share of that cost can be removed before the detector runs. We presented AnchorSlice, a static slicing stage that reduces a Solidity contract to the fragments capable of hosting each DASP category, and evaluated it on the DIVE benchmark by changing only the detector's input. This section summarises the findings and the directions they open.

AnchorSlice is lexical throughout, needs no compilation, and processes a contract in 7.6 ms on average. On a stratified sample of 50 contracts it retained evidence for all 119 labelled category instances while emitting 60.2% of the original characters. In a paired evaluation it reduced token consumption by 56.6% and detection runtime by 19.7%, while raising recall from 47.9% to 63.0%, F1-score from 42.7% to 50.0% and Cohen's kappa from 0.145 to 0.220. These detection gains are not statistically significant, and the results come from a single run of one detector on an in-sample set of anchors, but they indicate that taxonomy-aware input reduction is compatible with, and can benefit, source-level vulnerability detection.

Four directions follow. A tag-free arm would separate the effect of removing code from the effect of naming candidate categories, and would also remove the header overhead that currently offsets the anchoring stage. A selectivity threshold that suppresses any category tagging more than a set fraction of a contract's blocks would address the DoS failure directly while preserving the access-control gain. The anchors should be re-validated on a held-out DIVE split, with the protocol repeated across several runs, before the approach is applied to the full corpus. Finally, the slice should be evaluated in front of other detectors, including open-weight models and our LineGuard framework, to establish that the token savings transfer across detection architectures.

VII. DISCLOSURE STATEMENT

The authors declare that they have no relevant or material financial interests related to the research described in this paper.

REFERENCES

- [1] Szabo, N.: 'Formalizing and Securing Relationships on Public Networks', *First Monday*, 1997, 2, (9), doi: [10.5210/fm.v2i9.548](https://doi.org/10.5210/fm.v2i9.548)
- [2] Wood, G.: 'Ethereum: A Secure Decentralised Generalised Transaction Ledger', *Ethereum Project Yellow Paper*, 2014, <https://ethereum.github.io/yellowpaper/paper.pdf>
- [3] Luu, L., Chu, D.-H., Olickel, H., Saxena, P., Hobor, A.: 'Making Smart Contracts Smarter'. *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS)*, 2016, pp. 254–269, doi: [10.1145/2976749.2978309](https://doi.org/10.1145/2976749.2978309)
- [4] NCC Group: 'Decentralized Application Security Project (DASP) Top 10', 2018, <https://dasp.co>
- [5] Durieux, T., Ferreira, J.F., Abreu, R., Cruz, P.: 'Empirical Review of Automated Analysis Tools on 47,587 Ethereum Smart Contracts'. *Proc. ACM/IEEE Int. Conf. Software Engineering (ICSE)*, 2020, pp. 530–541, doi: [10.1145/3377811.3380364](https://doi.org/10.1145/3377811.3380364)
- [6] Ghaleb, A., Pattabiraman, K.: 'How Effective Are Smart Contract Analysis Tools? Evaluating Smart Contract Static Analysis Tools Using Bug Injection'. *Proc. ACM SIGSOFT Int. Symp. Software Testing and Analysis (ISSTA)*, 2020, pp. 415–427, doi: [10.1145/3395363.3397385](https://doi.org/10.1145/3395363.3397385)
- [7] Feist, J., Grieco, G., Groce, A.: 'Slither: A Static Analysis Framework for Smart Contracts'. *Proc. IEEE/ACM Int. Workshop on Emerging Trends in Software Engineering for Blockchain (WETSEB)*, 2019, pp. 8–15, doi: [10.1109/WETSEB.2019.00008](https://doi.org/10.1109/WETSEB.2019.00008)
- [8] Zhuang, Y., Liu, Z., Qian, P., Liu, Q., Wang, X., He, Q.: 'Smart Contract Vulnerability Detection Using Graph Neural Network'. *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2020, pp. 3283–3290, doi: [10.24963/ijcai.2020/454](https://doi.org/10.24963/ijcai.2020/454)
- [9] Sun, Y., Wu, D., Xue, Y., Liu, H., Wang, H., Xu, Z., Xie, X., Liu, Y.: 'GPTScan: Detecting Logic Vulnerabilities in Smart Contracts by Combining GPT with Program Analysis'. *Proc. IEEE/ACM Int. Conf. Software Engineering (ICSE)*, 2024, pp. 1–13, doi: [10.1145/3597503.3639117](https://doi.org/10.1145/3597503.3639117)
- [10] Hu, S., Huang, T., İlhan, F., Tekin, S.F., Liu, L.: 'Large Language Model-Powered Smart Contract Vulnerability Detection: New Perspectives'. *Proc. IEEE Int. Conf. Trust, Privacy and Security in Intelligent Systems and Applications (TPS-ISA)*, 2023, pp. 297–306, doi: [10.1109/TPS-ISA58951.2023.00044](https://doi.org/10.1109/TPS-ISA58951.2023.00044)
- [11] Ma, W., Wu, D., Sun, Y., Wang, T., Liu, S., Zhang, J., Xue, Y., Liu, Y.: 'Combining Fine-Tuning and LLM-Based Agents for Intuitive Smart Contract Auditing with Justifications'. *Proc. IEEE/ACM Int. Conf. Software Engineering (ICSE)*, 2025, pp. 1–13, [arXiv:2403.16073](https://arxiv.org/abs/2403.16073)
- [12] He, Z., Li, Z., Yang, S., Qiao, A., Zhang, X., Luo, X., Chen, T.: 'Large Language Models for Blockchain Security: A Systematic Literature Review', *arXiv preprint*, 2024, [arXiv:2403.14280](https://arxiv.org/abs/2403.14280)
- [13] Liu, N.F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., Liang, P.: 'Lost in the Middle: How Language Models Use Long Contexts', *Trans. Assoc. Computational Linguistics*, 2024, 12, pp. 157–173, doi: [10.1162/tacl_a_00638](https://doi.org/10.1162/tacl_a_00638)
- [14] Jiang, H., Wu, Q., Lin, C.-Y., Yang, Y., Qiu, L.: 'LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models'. *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 13358–13376, doi: [10.18653/v1/2023.emnlp-main.825](https://doi.org/10.18653/v1/2023.emnlp-main.825)
- [15] Weiser, M.: 'Program Slicing'. *Proc. Int. Conf. Software Engineering (ICSE)*, 1981, pp. 439–449, <https://doi.org/10.5555/800078.802557>
- [16] Pakshad, P., Aghanasiri, S., Shamel-Sendi, A., Omar, M.: 'LineGuard: An LLM-Based Smart Contract Vulnerability Detection', manuscript under review, 2025, <https://github.com/sjag77/LineGuard>
- [17] Alsunaidi, S.J., Aljamaan, H., Hammoudeh, M.: 'DIVE: A Multi-Label Smart Contract Vulnerability Dataset', *Scientific Data*, 2026, 13, art. 664; dataset: [10.5281/zenodo.18519253](https://doi.org/10.5281/zenodo.18519253)
- [18] Tsankov, P., Dan, A., Drachler-Cohen, D., Gervais, A., Bünzli, F., Vechev, M.: 'Securify: Practical Security Analysis of Smart Contracts'. *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS)*, 2018, pp. 67–82, doi: [10.1145/3243734.3243780](https://doi.org/10.1145/3243734.3243780)
- [19] Tikhomirov, S., Voskresenskaya, E., Ivanitskiy, I., Takhaviev, R., Marchenko, E., Alexandrov, Y.: 'SmartCheck: Static Analysis of Ethereum Smart Contracts'. *Proc. IEEE/ACM Int. Workshop on Emerging Trends in Software Engineering for Blockchain (WETSEB)*, 2018, pp. 9–16, doi: [10.1145/3194113.3194115](https://doi.org/10.1145/3194113.3194115)
- [20] ConsenSys: 'Mythril: Security Analysis Tool for EVM Bytecode', 2018, <https://github.com/ConsenSys/mythril>
- [21] Nikolić, I., Kolluri, A., Sergey, I., Saxena, P., Hobor, A.: 'Finding the Greedy, Prodigal, and Suicidal Contracts at Scale'. *Proc. Annual Computer Security Applications Conf. (ACSAC)*, 2018, pp. 653–663, doi: [10.1145/3274694.3274743](https://doi.org/10.1145/3274694.3274743)
- [22] So, S., Lee, M., Park, J., Lee, H., Oh, H.: 'VeriSmart: A Highly Precise Safety Verifier for Ethereum Smart Contracts'. *Proc. IEEE*

- Symp. Security and Privacy (S&P)*, 2020, pp. 1678–1694, doi: [10.1109/SP40000.2020.00032](https://doi.org/10.1109/SP40000.2020.00032)
- [23] Jiang, B., Liu, Y., Chan, W.K.: ‘ContractFuzzer: Fuzzing Smart Contracts for Vulnerability Detection’. *Proc. ACM/IEEE Int. Conf. Automated Software Engineering (ASE)*, 2018, pp. 259–269, doi: [10.1145/3238147.3238177](https://doi.org/10.1145/3238147.3238177)
- [24] Nguyen, T.D., Pham, L.H., Sun, J., Lin, Y., Minh, Q.T.: ‘sFuzz: An Efficient Adaptive Fuzzer for Solidity Smart Contracts’. *Proc. ACM/IEEE Int. Conf. Software Engineering (ICSE)*, 2020, pp. 778–788, doi: [10.1145/3377811.3380334](https://doi.org/10.1145/3377811.3380334)
- [25] Li, Z., Zou, D., Xu, S., Ou, X., Jin, H., Wang, S., Deng, Z., Zhong, Y.: ‘VulDeePecker: A Deep Learning-Based System for Vulnerability Detection’. *Proc. Network and Distributed System Security Symp. (NDSS)*, 2018, doi: [10.14722/ndss.2018.23158](https://doi.org/10.14722/ndss.2018.23158)
- [26] Li, Z., Zou, D., Xu, S., Jin, H., Zhu, Y., Chen, Z.: ‘SySeVR: A Framework for Using Deep Learning to Detect Software Vulnerabilities’, *IEEE Trans. Dependable and Secure Computing*, 2022, 19, (4), pp. 2244–2258, doi: [10.1109/TDSC.2021.3051525](https://doi.org/10.1109/TDSC.2021.3051525)
- [27] Zhou, Y., Liu, S., Siow, J., Du, X., Liu, Y.: ‘Devign: Effective Vulnerability Identification by Learning Comprehensive Program Semantics via Graph Neural Networks’. *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2019, pp. 10197–10207, [arXiv:1909.03496](https://arxiv.org/abs/1909.03496)
- [28] Fu, M., Tantithamthavorn, C.: ‘LineVul: A Transformer-Based Line-Level Vulnerability Prediction’. *Proc. IEEE/ACM Int. Conf. Mining Software Repositories (MSR)*, 2022, pp. 608–620, doi: [10.1145/3524842.3528452](https://doi.org/10.1145/3524842.3528452)
- [29] Feng, Z., Guo, D., Tang, D., Duan, N., Feng, X., Gong, M., Shou, L., Qin, B., Liu, T., Jiang, D., Zhou, M.: ‘CodeBERT: A Pre-Trained Model for Programming and Natural Languages’. *Findings of the Association for Computational Linguistics: EMNLP*, 2020, pp. 1536–1547, doi: [10.18653/v1/2020.findings-emnlp.139](https://doi.org/10.18653/v1/2020.findings-emnlp.139)
- [30] Liu, Z., Qian, P., Wang, X., Zhuang, Y., Qiu, L., Wang, X.: ‘Combining Graph Neural Networks with Expert Knowledge for Smart Contract Vulnerability Detection’, *IEEE Trans. Knowledge and Data Engineering*, 2023, 35, (2), pp. 1296–1310, doi: [10.1109/TKDE.2021.3095196](https://doi.org/10.1109/TKDE.2021.3095196)
- [31] Chen, M., et al.: ‘Evaluating Large Language Models Trained on Code’, *arXiv preprint*, 2021, [arXiv:2107.03374](https://arxiv.org/abs/2107.03374)
- [32] Brown, T.B., et al.: ‘Language Models Are Few-Shot Learners’. *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2020, pp. 1877–1901, [arXiv:2005.14165](https://arxiv.org/abs/2005.14165)
- [33] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., Zhou, D.: ‘Chain-of-Thought Prompting Elicits Reasoning in Large Language Models’. *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2022, pp. 24824–24837, [arXiv:2201.11903](https://arxiv.org/abs/2201.11903)
- [34] Sun, Y., Wu, D., Xue, Y., Liu, H., Ma, W., Zhang, L., Shi, M., Liu, Y.: ‘LLM4Vuln: A Unified Evaluation Framework for Decoupling and Enhancing LLMs’ Vulnerability Reasoning’, *arXiv preprint*, 2024, [arXiv:2401.16185](https://arxiv.org/abs/2401.16185)
- [35] Wei, Z., Sun, J., Zhang, Z., Zhang, X., Yang, X., Zhu, L.: ‘LLM-SmartAudit: Advanced Smart Contract Vulnerability Detection’, *arXiv preprint*, 2024, [arXiv:2410.09381](https://arxiv.org/abs/2410.09381)
- [36] Yu, L., Chen, S., Yuan, H., Wang, P., Huang, Z., Zhang, J., Shen, C., Zhang, F., Yang, J., Li, J.: ‘Smart-LLaMA: Two-Stage Post-Training of Large Language Models for Smart Contract Vulnerability Detection and Explanation’, *arXiv preprint*, 2024, [arXiv:2411.06221](https://arxiv.org/abs/2411.06221)
- [37] Liu, Y., Xue, Y., Wu, D., Sun, Y., Li, Y., Shi, M., Liu, Y.: ‘PropertyGPT: LLM-Driven Formal Verification of Smart Contracts Through Retrieval-Augmented Property Generation’. *Proc. Network and Distributed System Security Symp. (NDSS)*, 2025, doi: [10.14722/ndss.2025.241357](https://doi.org/10.14722/ndss.2025.241357)
- [38] Jiang, H., Wu, Q., Luo, X., Li, D., Lin, C.-Y., Yang, Y., Qiu, L.: ‘LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression’. *Proc. Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024, pp. 1658–1677, doi: [10.18653/v1/2024.acl-long.91](https://doi.org/10.18653/v1/2024.acl-long.91)
- [39] Lewis, P., et al.: ‘Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks’. *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2020, pp. 9459–9474, [arXiv:2005.11401](https://arxiv.org/abs/2005.11401)
- [40] Sridharan, M., Fink, S.J., Bodík, R.: ‘Thin Slicing’. *Proc. ACM SIGPLAN Conf. Programming Language Design and Implementation (PLDI)*, 2007, pp. 112–122, doi: [10.1145/1250734.1250748](https://doi.org/10.1145/1250734.1250748)
- [41] Yashavant, C.S., Kumar, S., Karkare, A.: ‘ScrawlID: A Dataset of Real World Ethereum Smart Contracts Labelled with Vulnerabilities’, *arXiv preprint*, 2022, [arXiv:2202.11409](https://arxiv.org/abs/2202.11409)
- [42] Zheng, Z., Su, J., Chen, J., Lo, D., Zhong, Z., Ye, M.: ‘DAppSCAN: Building Large-Scale Datasets for Smart Contract Weaknesses in DApp Projects’, *IEEE Trans. Software Engineering*, 2024, 50, (6), pp. 1360–1373, doi: [10.1109/TSE.2024.3383422](https://doi.org/10.1109/TSE.2024.3383422)
- [43] Repository available at <https://github.com/sjag77/AnchorSlice>



SAJAD AGHANASIRI is currently an M.Sc. student in Software Engineering at Shahid Beheshti University (SBU), Tehran, Iran. He received his B.Sc. in Computer Engineering from Islamic Azad University, Central Tehran Branch (IAU-CTB). His thesis focuses on securing smart contracts via vulnerability detection in code snippets and program analysis, exploring static, semantic, and hybrid techniques. His research interests span software security, blockchain/smart-contract security, program analysis, and complex networks. He has served as a Teaching Assistant for one academic term.

SINA ZAKER [Biography to be provided.]



ALIREZA SHAMELI-SENDI is currently an Associate Professor at Shahid Beheshti University. Before joining SBU, he was a Postdoctoral Fellow at Ericsson, Canada and Postdoctoral at ETS and McGill universities in collaboration with Ericsson. He received his Ph.D degree in computer engineering from Montreal University (Ecole Polytechnique de Montreal), Canada. He obtained his B.Sc. and M.Sc. from Amirkabir University of Technology. His primary research interests include information security, intrusion response system, and cloud computing. He is a recipient of Postdoctoral Research Fellowship Award and Industrial Postdoctoral Fellowship Award from Canada. In addition, he received the best researcher awards, at SBU, in 2018 and 2022.

...